

# Choosing a gauge

A decision report for the gauge that replaces tree–cotree, written against the primary sources rather than summaries of them.

**Second edition.** The first draft recommended Zhao & Fu's dummy-variable Coulomb gauge on cost-and-fit grounds. That recommendation is **withdrawn**. It was made without the potential-BEM endgame in view and before the primary sources had been obtained; both change the answer.

## 0. Decision

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended gauge         | Chew's generalized gauge — $\nabla \cdot (\epsilon \mathbf{A}) = -\chi \partial \Phi / \partial t$ , $\chi = \alpha \epsilon^2 \mu$                                                                                                                                                                                                                                                                                              |
| Why, in one line          | It is the only candidate under which the FEM half, the interface conditions, and the BEM half all come from one formulation.                                                                                                                                                                                                                                                                                                     |
| Is anything broken today? | <b>Not for any geometry in the suite — and every one of them puts its terminals on the outer boundary.</b> $\mathbf{E}$ , $\mathbf{B}$ , $\mathbf{H}$ , $\mathbf{J}$ , terminal $\mathbf{V}$ , $\mathbf{I}$ , $\mathbf{R}$ , $\mathbf{L}$ , $\mathbf{Z}$ are correct and gauge-invariant there, measured to 1e-12 by <code>07_GaugeInvariance</code> . That result does <b>not</b> extend to a port inside the domain: see §1.2. |
| What is wrong             | $\Phi$ in the interior is not physical; tree–cotree postpones rather than removes low-frequency breakdown; and <b>at a port face inside the domain — especially one crossing conductor and dielectric — it has no principled rule at all, at any frequency.</b>                                                                                                                                                                  |
| What it costs us          | Less than the first edition assumed: <b>**our Whitney-1 <math>\mathbf{A}</math> space is already the one Chew's gauge requires.**</b> $\Phi$ moves $P2 \rightarrow P1$ and the gauge term is added; the $\mathbf{A}$ discretization is untouched.                                                                                                                                                                                |
| Still open                | Whether $\mathbf{A}$ stays a Whitney 1-form <b>at the boundary-integral surface</b> . This is now the real research question — see §6.                                                                                                                                                                                                                                                                                           |
| Who decides               | The mixed-port benchmark the roadmap requires. This report narrows the field to one candidate; it does not retire the benchmark.                                                                                                                                                                                                                                                                                                 |

# 1. Where we stand

Tree–cotree sets  $\mathbf{A} = \mathbf{0}$  on a spanning tree's edges. That makes the matrix non-singular and nothing more. `07_GaugeInvariance` measures the consequence on one problem solved on three different trees:

|                                                  | CHANGE ACROSS TREES  |
|--------------------------------------------------|----------------------|
| $\Phi, \mathbf{A}$                               | 0.14 – 0.22 relative |
| $\mathbf{E}, \mathbf{B}, \mathbf{H}, \mathbf{J}$ | 4.7e-13 ... 3.2e-12  |
| terminal $\mathbf{V}, \mathbf{R}, \mathbf{L}$    | 2.7e-13              |

`docs/FIELD_POSTPROCESSING.md` §9 quantifies how much of  $\Phi$  is gauge: about  $\omega L / |Z| = \sin(\arg Z)$ . Case 06 sits at  $\omega L / R = 4.48$ , case 04 at 12.8 — both gauge-dominated, which is why neither asserts  $\Phi$ .

**A consequence worth stating before anything else.** Our terminal quantities are gauge-independent *only* because  $\mathbf{n} \times \mathbf{A} = \mathbf{0}$  on the outer boundary forces the gauge function  $\psi$  to be constant there — measured,  $|\psi|$  is 3.6e-16 on the boundary against 4.6e-04 inside.  $\mathbf{n} \times \mathbf{A} = \mathbf{0}$  is a condition on  $\mathbf{A}$ , so it is not gauge-invariant; that is exactly why it pins  $\psi$ .

**Replace it with a radiation, transmission or boundary-integral condition — all of which are written in the gauge-invariant fields — and  $\psi$  stops being pinned at all.** The moment the boundary treatment becomes correct for THz, tree–cotree leaves not only interior  $\Phi$  but the terminal quantities floating. The gauge must become a genuine PDE condition *before* the boundary condition changes, not after.

## 1.1 Why tree–cotree's $\Phi$ has no continuum limit — Rapetti et al. settle it

F. Rapetti, A. Alonso Rodríguez and E. De Los Santos, *On the Tree Gauge in Magnetostatics*, **J 5(1), 52–63, 2022**, DOI 10.3390/j5010004.

The project's proposal attributes to this paper the claim that the tree gauge is not a discretization of any orthogonality condition. **The claim is correct, and the paper states it outright.** Their §4, immediately after defining the tree gauge  $\mathbf{a}_t = \mathbf{0}$ :

It is worth noting that **these tree gauges are not a discretization of the Coulomb gauge** stated in (5) or (6). **They are not enforcing in any sense the orthogonality to the gradient** in conditions (5) and (6).

And their §5 opens by saying the tree degrees of freedom

**is set arbitrarily**, eventually equal to zero, without affecting the corresponding field  $B_h = \text{curl } A_h$ .

That is case 07's result stated in the literature — for  $A$ , in magnetostatics, from the algebraic side rather than by measurement.

**This is the deepest available answer to "why is tree-cotree's  $\Phi$  not a physical quantity."**

The discrete Coulomb gauge *is* an orthogonality condition — their eq. (11) says  $A_h$  is orthogonal to  $\text{grad}(W^0)$ , and their §5 constructs it as a projection onto  $(\ker S)^\perp = \text{Im}(S)$ . The tree gauge is not the discretization of anything in the continuum; it is a choice of which algebraic degrees of freedom to delete. A quantity fixed by such a choice **cannot converge to a definite function under mesh refinement, because there is no continuum condition for it to converge to.**

That is the structural reason behind every measurement in this report, and it is why §9's first test — pointwise mesh convergence of  $\Phi$  — is the one that matters.

## 1.2 The limit of what case 07 proves — interior and mixed ports

Case 07's result is **conditional on where the ports are**, and the condition is easy to lose sight of because every case in the suite satisfies it.

The terminal quantities are gauge-independent because  $n \times A = 0$  on the outer boundary forces  $\psi$  constant there and a port pins it to zero. **Both terminals of every case in this suite lie in the outer boundary** — the conductors span the full height of their air boxes, by construction. Measured:  $|\psi|$  is  $3.6\text{e-}16$  on the boundary against  $4.6\text{e-}04$  in the interior, twelve orders apart.

**A port face inside the domain has none of that protection.**  $\psi$  is unpinned there, so  $\Phi$  on such a face is gauge-dependent — which matters not only for reading a voltage out but for *posing* a current-driven port, whose terminal potential is a free unknown.

A face crossing **conductor and dielectric** is worse than merely interior. The tree gauge grounds tree paths to conductors; on a face that is partly conductor and partly dielectric there is no principled rule for which edges to take, and — by Rapetti et al. (§1.1) — **no continuum condition to appeal to**, because the tree gauge is not a discretization of one. There is nothing to be right or wrong against.

**This is not a high-frequency problem.** Ansari et al. (§3.1) demonstrate non-unique potentials in the **quasi-static** regime: a conductive prism in a conductive background, driven by a grounded wire — a galvanic low-frequency geophysics problem with material contrasts inside the domain. Their fields were fine and their potentials were not. **The failure mode is about material contrast and port placement, not about frequency.**

**What this project has and has not measured.** We have measured, thoroughly, that tree–cotree is sound for boundary-mounted ports. **We have never built a case with an interior or mixed-material port, so we have not measured the failure either** — it is predicted by the mechanism above and documented in the literature, not yet observed here. §9 proposes closing that gap, and it should be closed *before* the gauge change, so that there is a failing baseline to compare against.

### 1.3 Measured: an interior cut's LOCAL field is tree-dependent at 8 %

The strongest evidence in this report is not from the literature. It is a measurement on **05\_Loop\_1A\_50Hz**, a case that **passes all its assertions**, at **50 Hz**, through a cut that lies **entirely inside a conductor** — the most favourable interior-port configuration there is, with one side of the cut even pinned at 0 V.

Case 05's ring is driven by an internal cut and asserts  **$L_{nH} = 19.7854 \pm 0.05$** , obtained from  **$Z = V / I_{drive}$** , so the assertion rides on a potential. Running **07\_GaugeInvariance**'s permutation harness on it, three spanning trees:

|                                           | BASE → PERMA      | BASE → PERMB |                |                |
|-------------------------------------------|-------------------|--------------|----------------|----------------|
| <b>L</b> (asserted $19.7854 \pm 0.05$ nH) | <b>+0.0035 nH</b> | +0.0009 nH   |                |                |
| <b>\</b>                                  | <b>V\</b>         | <b>`</b>     | ~3e-6 relative | ~3e-6 relative |
| <b>** J</b> within 2 mm of the cut**      | <b>8.3 %</b>      | —            |                |                |
| <b>** B</b> within 2 mm of the cut**      | <b>7.4 %</b>      | —            |                |                |

and the field change decays sharply away from the cut — **J** goes **8.3 % → 2.3e-4 → 7.2e-6** across the 0–2, 2–5 and 5–10 mm bands. It is localized at the port, not global.

**Three checks say it is real rather than numerical noise:**

- all three solves are exact — backward errors 9.1e-23, 9.0e-23, 9.2e-23;
- $\Delta E = \Delta J / \sigma$**  to a ratio of **1.0000**, so the **E** and **J** changes are one

physical change reported against different global maxima;

- it decays smoothly over ~11,000 nodes, so it is not the 205 cut nodes being mislabelled in the output.

**The integrated quantities survive and the local ones do not.**  $L$  moves by 0.018 %, comfortably inside a tolerance it was never in danger of failing — which is exactly why case 05 passes and why nothing flagged this. **Case 05 is not wrong.** What is wrong is that the current distribution within a couple of millimetres of the feed depends on an arbitrary spanning tree, and current distribution near a feed is precisely what a phased-array analysis is for.

**Practitioner experience, recorded as corroboration.** From the project owner's own use of a commercial A- $\Phi$  solver: a mixed-material port ran to completion without crashing and returned wrong solutions, and a Coulomb-type gauge improved matters without fully fixing them. The working assumption there was that **internal ports lying inside a conductor were fine under tree-cotree, and only mixed-material interior ports were problematic.**

(Observed user-visible behaviour only; nothing here is a claim about that tool's internals.)

**This measurement refines that assumption rather than contradicting it.** For the quantities such a port is normally asked for — terminal voltage, R, L, Z — the assumption holds, and holds well: 0.018 % on  $L$ . It is the **local field** that moves, by 8 %, and no integrated check would ever reveal it. So the conductor-only interior port is not "fine"; it is **fine for circuit quantities and unreliable for field quantities**, and the distinction has probably gone unnoticed because the things people assert about such ports are all integrals.

**Why this is the report's most useful result.** It is ours, it is measured, it is at low frequency, and it is on a case that passes — so it cannot be dismissed as a pathological configuration or a high-frequency effect. It also sets the bar for the replacement: a new gauge has to make the local field near an interior port tree-independent, not merely keep  $L$  inside tolerance.

---

## 2. Chew 2014 is the keystone

W. C. Chew, *Vector Potential Electromagnetics with Generalized Gauge for Inhomogeneous Media: Formulation* (Invited Paper), **Progress In Electromagnetics Research, Vol. 149, pp. 69–84, 2014**, DOI 10.2528/PIER14060904. Open access; now in [APhi\\_Papers/Chew2014\\_GeneralizedGauge\\_PIER149.pdf](#).

### 2.1 The gauge

His eq. (15), with (18):

$$\nabla \cdot (\epsilon \mathbf{A}) = -\chi \partial \Phi / \partial t \quad \chi = \alpha \epsilon^2 \mu$$

where  $\alpha$  may be a function of position.  $\alpha = 1$  recovers the generalized Lorenz gauge  $\epsilon^{-1} \nabla \cdot (\epsilon \mathbf{A}) = -\mu \epsilon \partial \Phi / \partial t$  (his eq. 14); a homogeneous medium with  $\alpha = 1$  recovers the ordinary Lorenz gauge.

**\*\*  $\chi = \alpha \epsilon^2 \mu$  is confirmed from the source.\*\*** The first edition of this report flagged the roadmap's rendering as unverified because it had been read out of garbled PDF text. It is correct.

The resulting equations (16) and (17) decouple, and (16) is derivable from (17) by taking the divergence and using charge continuity. Chew states plainly that, unlike the vector wave equations for the fields, **these do not exhibit the low-frequency catastrophe**, so ordinary FEM/FD solvers work across scales and bandwidth.

He also rules out the alternative explicitly: under the **Coulomb gauge** the scalar potential has infinite velocity and *the vector potential equation is not completely decoupled from the scalar potential equation*. That single sentence disqualifies every Coulomb-type candidate for a potential-BEM hybrid, for the reason developed in §4.

### 2.2 It dissolves the port problem — which is why you went looking

§3, on the interface between two media:

the boundary conditions at the interface of two homogeneous media are **also embedded in these equations**. Therefore, when one solves the PDEs directly in an inhomogeneous medium, **one need not stipulate the boundary conditions**. The solutions naturally obey

the boundary conditions if they are arrived at correctly via a numerical method.

The conditions he then derives are the classical ones:

$$\hat{n} \times A_1 = \hat{n} \times A_2 \quad (23) \quad \text{tangential A continuous}$$

$$\hat{n} \times (1/\mu_1) \nabla \times A_1 = \hat{n} \times (1/\mu_2) \nabla \times A_2 \quad \equiv \quad \hat{n} \times H_1 = \hat{n} \times H_2 \quad (24), (25)$$

$$\hat{n} \cdot (\epsilon_1 A_1) = \hat{n} \cdot (\epsilon_2 A_2) \quad (26) \quad \text{normal } \epsilon A \text{ continuous}$$

$$\hat{n} \cdot (\epsilon_1 \nabla \Phi_1) = \hat{n} \cdot (\epsilon_2 \nabla \Phi_2) \quad (27)$$

together giving  $\hat{n} \cdot (\epsilon_1 E_1) = \hat{n} \cdot (\epsilon_2 E_2)$ , the usual normal- $D$  condition.

**Equation (26) is the one that decides our element choice**, and §6 is about why: it is  $\epsilon A$ , not  $A$ , whose normal component is continuous, so  $\hat{n} \cdot A$  jumps wherever  $\epsilon$  jumps.\*\*

**So a port face that crosses conductor and dielectric needs no special treatment.** The gauge carries the interface physics. [ROADMAP.md](#) Phase 03 identifies exactly this as the reason tree–cotree is inadequate; this is the paper that answers it.

He also treats the PEC limit: as  $\epsilon_2 \rightarrow \infty$ ,  $\hat{n} \times \nabla \Phi_2 = 0$ , so  $\Phi_2$  is constant in the PEC and  $\hat{n} \times E_1 = 0$  even at  $\omega = 0$ .

## 2.3 It already contains the boundary-integral half

§6 derives the **generalized Green's theorem, the extinction theorem and the surface equivalence principle** for the  $A$ - $\Phi$  formulation; §7 derives a **surface integral equation for a PEC scatterer** with its matrix representation. Per the abstract, those integral equations exhibit **neither the low-frequency catastrophe nor the frequency imbalance** of the classical  $E-H$  ones.

**This is the decisive architectural fact: the FEM half, the interface conditions, and the BEM half are one formulation under one gauge.** No other candidate offers that.

Two implementation details from §6–7 worth recording now:

- For the SIE derivation Chew assumes the **simple Lorenz gauge**, because SIEs

apply to piecewise-*homogeneous* regions where  $\epsilon$  is constant. The generalized gauge is what the inhomogeneous FEM volume needs; the simple one is what the homogeneous exterior needs. They agree where it matters.

- The equivalence principle needs **six scalar quantities** on the surface: two

tangential components of  $\mathbf{H}_1$ , two tangential components of  $\mathbf{A}_1$ ,  $\Phi$ , and  $\hat{\mathbf{n}} \cdot \mathbf{A}_1$ . He notes this parallels the augmented EPA, where six components are likewise needed to keep low-frequency stability.

- The surface current basis must be **divergence-conforming** (RWG-type) so that the  $\mathbf{A}$  it produces is divergence-conforming too.

### 3. A published statement of the problem case 07 measures

S. Sharma and P. Triverio, *Electromagnetic Modeling of Lossy Interconnects From DC to High Frequencies With a Potential-Based Boundary Element Formulation*, IEEE Trans. Electromagn. Compat. **64**(4), 2022 — arXiv:2112.07360.

They use the **usual Lorenz gauge outside** and a **modified Lorenz gauge inside** a lossy conductor:

$$\nabla \cdot \tilde{\mathbf{A}} = -j\omega\epsilon_0\mu_0 \Phi \quad \text{in free space} \quad (17)$$

$$\nabla \cdot \tilde{\mathbf{A}} = -(j\omega\epsilon + \sigma) \mu \Phi_r \quad \text{in the conductor} \quad (19)$$

$$\Phi_r = \Phi - \Phi_a, \quad \Phi_a = \text{average of } \Phi \text{ over the object's surface} \quad (20), (21)$$

Their stated reason is the one this project measured independently:

in order to determine  $\Phi$  uniquely within  $V$ , one must specify a boundary condition on  $S$  and a reference point; if the conventional Lorenz gauge is used, \*\*a change in reference for  $\Phi$  will lead to a change in  $\nabla \cdot \tilde{\mathbf{A}}|_V$  \*\* — in other words  $\nabla \cdot \tilde{\mathbf{A}}|_V$  cannot be determined uniquely until a reference is set for  $\Phi$ .

Subtracting the object's own average surface potential makes the interior gauge reference-independent, and both regions then yield clean Helmholtz equations.

**This is a close published analogue to case 07's result** — the potential *reference* contaminating the vector potential, where ours is the *tree* doing it. The disease is the same: a bookkeeping choice leaking into a solved quantity. It is worth citing in the case 07 README.

3.1 Ansari et al. is the closest analogue of all

M. Ansari, C. G. Farquharson and S. P. MacLachlan, *A gauged finite-element potential formulation for accurate inductive and galvanic modelling of 3-D electromagnetic problems*, **Geophys. J. Int.** **210(1)**, **105–129**, **2017**, DOI 10.1093/gji/ggx149. Paywalled; abstract and technical content read from the publisher's article page, PDF not yet obtained.

They demonstrate "**non-unique, incorrect potentials**" from the usual incompletely-gauged system — and the demonstration is the striking part. They show it by

showing **inconsistent results obtained from iterative and direct linear equation solvers**

— the same physical fields, but **different potentials depending on which linear solver was used**. After introducing the Coulomb gauge explicitly as an extra equation and augmenting the Helmholtz equation with the gradient of a Lagrange multiplier, "both the iterative and direct solvers produce the same responses for the potentials, demonstrating the uniqueness of the numerical solution."

Two further points make this the most directly transferable result in the collection:

- **\*\*They use edge elements for  $\mathbf{A}$  and nodal elements for  $\Phi$  — our exact discretization.\*\***

- They identify the root cause as the **\*\*normal component of  $\mathbf{A}$  being discontinuous across material interfaces\*\***, which edge elements permit and which an incompletely-gauged system therefore leaves unconstrained.

**Three independent triggers, one disease.** Collecting what the literature and this project have now each observed:

| OBSERVED BY                     | THE POTENTIALS CHANGE WITH...                     | FIELDS UNCHANGED? |
|---------------------------------|---------------------------------------------------|-------------------|
| Ansari et al. (2017)            | the <b>linear solver</b> (iterative vs direct)    | yes               |
| Li, Sun, Dai & Chew (2015)      | the <b>initial guess</b> of an iterative solve    | yes               |
| Sharma & Triverio (2022)        | the <b>potential reference</b> inside a conductor | yes               |
| <b>** 07_GaugeInvariance **</b> | the <b>spanning tree</b>                          | yes, to 1e-12     |

Each trigger is different and each is incidental; the common cause is that nothing in the formulation pins the gauge.

Note also that they *exploit* gauge freedom rather than merely tolerating it: they leverage the gauge invariance of the potentials to devise simpler boundary conditions for **A** than existing potential-integral-equation methods use.

## 4. Why a potential-BEM forces a Lorenz-type gauge

This is physics, independent of any citation.

In the **Lorenz** gauge the exterior potentials satisfy

$(\nabla^2 + k_o^2) \mathbf{A} = 0$

$(\nabla^2 + k_o^2) \Phi = 0$

so **every** Cartesian component of **A**, and **Φ**, satisfies the same scalar Helmholtz equation — which is exactly what admits the scalar Green's function **G<sub>o</sub>** and a boundary representation built from standard single- and double-layer operators.

In the **Coulomb** gauge this collapses: **Φ** satisfies Poisson's equation (instantaneous, no **k<sub>o</sub>**) and **A**'s equation retains a **∇(∂Φ/∂t)** coupling term — which is precisely Chew's objection in §2.1. The clean scalar-Green's-function BEM is simply unavailable.

Corroborated by every potential-BEM paper obtained:

| GAUGE USED             |                                                          |
|------------------------|----------------------------------------------------------|
| Chew 2014 §6–7         | simple Lorenz for the homogeneous regions                |
| Sharma & Triverio 2022 | Lorenz outside, modified Lorenz inside                   |
| Li, Fu & Shanker 2017  | Lorenz; decoupling is what gives low-frequency stability |
| Abdrabou & Gomez 2026  | Lorenz, stated in the abstract                           |

## 5. Do hybrids still need gauging? Abdrabou & Gomez answer it

A. Abdrabou and L. J. Gomez, *A Hybrid DEC-SIE Framework for Potential-Based Electromagnetic Analysis of Heterogeneous Media*, J. Comput. Phys. 2026 — arXiv:2507.02099.

**They use no tree–cotree and no discrete gauging at all.** The gauge is imposed in continuous form before discretization, and they represent  **$\mathbf{A}$**  as three discrete 0-forms (nodal scalars) rather than the conventional 1-form. Their own framing: this sidesteps gauge-freedom issues at the discrete level by treating the components as independent scalars that satisfy the Lorenz constraint identically.

**The null space you gauge away is a property of the 1-form / edge-element representation.** Drop it and there is nothing to gauge.

That is the honest answer to "are gauges still needed in a hybrid": **not if you abandon Whitney 1-forms for  $\mathbf{A}$** . If you keep them — and we do — then yes.

Their scalar reformulation reduces the surface integral operators **from fourteen to two** (single- and double-layer). The reduction is performed by expressing the problem in Cartesian components and their normal derivatives; **the Lorenz gauge is the necessary enabler**, because it is what makes each component satisfy the same decoupled scalar equation in the first place. Gauge and reformulation are two steps, not one.

**A constraint their method carries — and how they dispose of it.** The coupling surface  $\Gamma$  must lie entirely in **free space** for the component-wise continuity of  $\mathbf{A}$  and  $\Phi$  to hold naturally: "the surface  $\Gamma$  has to be only in free space." Their fix is a **buffer region**: the DEC domain is the inhomogeneous body *plus* a homogeneous free-space collar,  $\Omega_2 = \Omega_s \cup \Omega_a$ , so that  $\Gamma = \partial\Omega_2$  is embedded entirely in free space, which in their words greatly simplifies enforcing the boundary conditions.

**This is directly transferable.** A THz array on a substrate puts the boundary-integral surface outside the substrate with a free-space collar, rather than cutting through it. It costs volume elements in the buffer and removes the material-crossing problem at the BI surface entirely.

Their reported condition number stays on the order of  **$10^3$**  across the tests, including as  $k_0 a \rightarrow 0$ .

5.1 Their method does not cover our problem

The paper contains no conductivity. Permeability is  $\mu_0$  throughout, the only material variation is  $\epsilon(\mathbf{r})$ , and every validation case is dielectric: a sphere ( $\epsilon = 2.25$ ), a multilayered sphere, and dielectric elliptical cylinders on a dielectric slab. A text search of the full paper returns **zero** occurrences of conductivity, conductor, lossy or  $\sigma$ .

A THz phased-array feed is the opposite case — lossy metal with a skin depth, sharp conductor edges, and conductor/dielectric interfaces where their permittivity-gradient coupling term becomes a surface delta. **The 0-form route is not something we could adopt off the shelf**, and §6 explains why it is not merely a matter of porting it.

6. The gauge and the element space are not independent choices

This is the central finding of the second edition, and it reframes the question.

6.1 The two representations

|                              | WHITNEY 1-FORM                                                     | WHITNEY 0-FORM           |
|------------------------------|--------------------------------------------------------------------|--------------------------|
| FEM name                     | lowest-order Nédélec edge element                                  | P1 nodal (Lagrange)      |
| basis                        | $w_{ij} = \lambda_i \nabla \lambda_j - \lambda_j \nabla \lambda_i$ | hat function $\lambda_i$ |
| degree of freedom            | $\int_{\text{edge}} \mathbf{A} \cdot d\mathbf{l}$ , one per edge   | the value at a node      |
| continuity enforced          | tangential only                                                    | full                     |
| what we use for $\mathbf{A}$ | ✓ this one                                                         | —                        |

In DEC the correspondence is exact: a discrete 1-form's DOF is the integral along a primal edge — the Nédélec edge DOF — and a 0-form's DOF is the nodal value.

6.2 Abdrabou & Gomez did not use Nédélec elements — they avoided them

In their own words: \*instead of modelling  $\mathbf{A}$  on the boundary as a 1-form, they treat its Cartesian components  $A_x, A_y, A_z$  as independent 0-forms\*, which they acknowledge departs from the standard DEC formulation. Their algorithm's output is literally **discrete 0-forms  $a_x, a_y, a_z, \phi_s$** .

In finite-element terms that is **node-based vector elements** — three copies of P1 Lagrange — the construction electromagnetic FEM normally forbids, for three reasons: spurious modes in curl–curl eigenproblems, inability to represent the singular field at a reentrant corner, and **being wrong at material interfaces**, where full continuity is imposed on a quantity whose normal component physically jumps.

### 6.3 Why they get away with it — the gauge decides

The third objection is the decisive one, and **which gauge you choose determines whether it applies**.

Chew generalized gauge
 $\nabla \cdot (\epsilon A) = -\chi \partial \Phi / \partial t$

$\Rightarrow \nabla \cdot (\epsilon A)$  finite at an interface

$\Rightarrow \hat{n} \cdot (\epsilon_1 A_1) = \hat{n} \cdot (\epsilon_2 A_2)$ 
Chew eq. (26)

$\Rightarrow \hat{n} \cdot A$  JUMPS wherever  $\epsilon$  jumps

$\Rightarrow A$  must live in  $H(\text{curl})$ : Whitney 1-form / Nédélec

Abdrabou gauge
 $\nabla \cdot A = i k_0^2 \epsilon(r) \Phi$

$\Rightarrow \nabla \cdot A$  finite at an interface

$\Rightarrow \hat{n} \cdot A$  is CONTINUOUS

$\Rightarrow A$  may live in  $H^1$ : nodal 0-forms are legitimate

Coulomb gauge
 $\nabla \cdot A = 0$

$\Rightarrow A$  solenoidal,  $\hat{n} \cdot A$  CONTINUOUS

$\Rightarrow$  nodal-natural; Ansari cites Biro & Preis for exactly this

**So the gauge and the basis cannot be chosen independently.** That is the real content of "1-form vs 0-form", and it is not stated anywhere in the project's existing documents.

| GAUGE                                                                                | $\hat{n} \cdot A$ AT A MATERIAL INTERFACE | ELEMENT SPACE IT SUITS |
|--------------------------------------------------------------------------------------|-------------------------------------------|------------------------|
| Coulomb $\nabla \cdot A = 0$                                                         | continuous                                | nodal                  |
| Abdrabou Lorenz $\nabla \cdot A = i k_0^2 \epsilon \Phi$                             | continuous                                | nodal                  |
| **Chew generalized $\nabla \cdot (\epsilon A) = -\chi \partial \Phi / \partial t$ ** | jumps $\propto 1/\epsilon$                | edge / Whitney-1       |

$A$  is gauge-dependent, so its *continuity class* is gauge-dependent too. There is no interface behaviour of  $A$  that is true independently of the gauge — only a behaviour consistent with each.

#### 6.3.1 This explains why Ansari needed a Lagrange multiplier — and why we would not

Ansari's diagnosis (§3.1) now reads as a **mismatch between gauge and element space**, not as a defect of either:

- He wanted the **Coulomb** gauge, under which  $\hat{n} \cdot \mathbf{A}$  must be **continuous**.
- He discretized  $\mathbf{A}$  with **edge elements**, which permit  $\hat{n} \cdot \mathbf{A}$  to **jump** — in

his words, the normal component of an edge-element vector potential "is not necessarily continuous across the boundary interfaces", and linear edge basis functions "do not necessarily form a solenoidal vector potential."

- So nothing in the discretization enforced what the gauge required, and the

potentials came out non-unique. The Lagrange multiplier is the machinery that repairs the mismatch.

**Chew's generalized gauge has no such mismatch.** It *wants*  $\hat{n} \cdot \mathbf{A}$  to jump at an  $\epsilon$  discontinuity, which is precisely what edge elements naturally permit. The gauge and our existing element space agree by construction rather than by enforcement.

That is an argument for Chew over every Coulomb-family candidate that is independent of the BEM argument in §4 — and it applies even if the hybrid never gets built.

Their resulting system makes the consequence visible: each component gets a plain scalar Helmholtz operator  $\nabla^2 \tilde{A}_v + k_0^2 \epsilon \tilde{A}_v$ , and the **only** coupling between components and  $\Phi$  is through  $\partial_v \epsilon$ , the gradient of permittivity. In a homogeneous region the three components decouple into independent scalar problems — which is exactly the regime nodal elements are built for.

### 6.4 What this means for us

**\*\*Our existing  $\mathbf{A}$  space is already the one Chew's gauge requires.\*\*** The first edition under-weighted this. Adopting Chew's generalized gauge means:

|                                             |                                                             |
|---------------------------------------------|-------------------------------------------------------------|
| $\mathbf{A}$ — Whitney 1-form edge elements | <b>unchanged</b>                                            |
| $\Phi$ — P2 nodal                           | → P1, to sit in Whitney-0 and restore the de Rham structure |
| the gauge term in the $\mathbf{A}$ equation | new: needs the Hodge/Whitney-mapping machinery              |
| tree–cotree                                 | removed                                                     |

Switching instead to the 0-form route would mean **\*\*replacing the  $\mathbf{A}$  discretization entirely, adopting a formulation with no demonstrated conductor support\*\*** (§5.1), and accepting nodal elements at the sharp metal edges of a THz feed — where objection (2) above still stands, gauge or no gauge.

## 6.5 The open question, stated precisely

Not *which gauge* — that is settled — but **\*\*whether  $\mathbf{A}$  stays a Whitney 1-form at the boundary-integral surface.\*\***

The fourteen operators arise *specifically* from the 1-form trace at the boundary. Abdrabou & Gomez's answer was to stop using it. Coupling Whitney 1-form  $\mathbf{A}$  to a potential BEM means taking on exactly the problem they chose to avoid: **that is the project's novelty claim and its principal technical risk, and the proposal should say so** rather than treating the fourteen-operator trap as something the gauge disposes of.

Chew's own §6 suggests a middle path worth evaluating before committing: his equivalence principle needs **six scalar surface quantities**, not fourteen operators — two tangential components of  $\mathbf{H}_1$ , two of  $\mathbf{A}_1$ ,  $\Phi$  and  $\hat{\mathbf{n}} \cdot \mathbf{A}_1$  — and his surface currents are **divergence-conforming**, which is far closer to Whitney-compatible than a Cartesian 0-form decomposition.

---

## 7. Caveats, including two from Chew himself

**\*\*Chew prefers  $\mathbf{E} - \mathbf{H}$  in genuinely wave-dominated regimes.\*\*** §5:

An advantage of (41) and (42) is that they are derivable from each other, whereas for the  $\mathbf{A}$ - $\Phi$  formulation, **only the  $\Phi$  equation is derivable from the  $\mathbf{A}$  equation**. Hence, when the  $\mathbf{E}$  and the  $\mathbf{H}$  fields are equally strong, and strongly coupled to each other, the  $\mathbf{E} - \mathbf{H}$  formulation is preferred, as it describes wave physics better.

This does not sink a THz solver — a phased-array feed is a multiscale problem where local low-frequency breakdown is the binding constraint — but **the  $\mathbf{A}$ - $\Phi$  advantage is multiscale robustness, not superior wave physics**, and the proposal should claim the former rather than the latter.

**\*\*The low-frequency *inaccuracy* problem is distinct from breakdown.\*\*** At  $\omega = 0$  the electrostatic solution lives in (16) and the magnetostatic solution in (17), so **both must be solved in tandem** to retrieve both accurately. Chew repeats this for the integral equations: the vector-potential SIE captures magnetoquasistatics, and a *scalar-potential* SIE is needed alongside it for electroquasistatics. Budget for solving both, not one.

**Topology.** At  $\omega = 0$  a PEC object of genus  $> 0$  admits non-trivial magnetostatic solutions with no excitation — superconducting loops. Our case 05 is a ring, so this is live for us, not hypothetical.

**What is still not verified.** Equations here were read from text extracted out of PDF content streams; the prose is reliable, the typeset mathematics is not. Every equation above should be checked against the typeset paper before it is implemented.

**All cited papers have now been obtained and read.** Ansari and Rapetti were supplied directly and are in [APhi\\_Papers/](#) ; everything else was downloaded from open-access sources. Nothing in this report now rests on a summary of a paper rather than the paper.

## 8. Candidates, final ranking

| VERDICT                                     |                                                                                                                                                                                                                                                                                      |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chew generalized gauge                      | <b>Recommended.</b> Only candidate where FEM, interface conditions and BEM are one formulation — and <b>our <math>A</math> space</b> is already the one it requires** (§6.4). Costs: Hodge/Whitney-mapping machinery, a sparse approximate inverse, and moving $\Phi$ from P2 to P1. |
| Zhao & Fu dummy-variable Coulomb            | <b>Withdrawn.</b> Symmetric, direct-solvable and a good architectural fit — but Coulomb-gauged, so it cannot couple to a potential BEM. Implementing it means implementing a gauge twice.                                                                                            |
| Ansari explicit Lagrange-multiplier Coulomb | Same Coulomb objection, plus a specific one: it exists to repair a <b>gauge/element mismatch we would not have</b> (§6.3.1). Retain as the roadmap's benchmark comparison — it is the documented failure mode a new gauge must not reproduce.                                        |
| Improve tree–cotree (root choice, MOR)      | Cheap, treats the symptom only. Lee & Jin themselves report the condition number still grows at low frequency.                                                                                                                                                                       |
| Do nothing                                  | Viable <i>only</i> while every port sits on an $n \times A = 0$ outer boundary. That excludes interior and mixed conductor/dielectric ports <b>at any frequency</b> (§1.2), and ends entirely when the boundary treatment changes (§1).                                              |

**A design decision the roadmap does not currently mention:** Chew's construction places  $\Phi$  in the Whitney-0 space, i.e. **P1 nodal. Ours is P2.** The compatibility argument rests on the Whitney/de Rham structure, so adopting the method means either moving  $\Phi$  to P1 or redoing the analysis for P2.

That is the *whole* discretization change. **A** stays exactly as it is — which, given that  $\hat{n} \cdot \mathbf{A}$  must be free to jump under this gauge (§6.3), is not a coincidence but a consequence of having picked edge elements correctly in the first place.

---

## 9. What to measure before committing

1. **Is  $\Phi$  mesh-independent?** Refine twice;  $\Phi$  must converge pointwise. Tree–cotree fails this by construction. **Case 07's permutation harness is directly reusable** — under a real gauge, its *potential-moves* assertions should invert while every field assertion holds unchanged.
2. **Condition number against frequency**, to the static limit, against the tree–cotree baseline.
3. **The mixed conductor/dielectric port**, with Chew's embedded interface conditions — does the port behave with no special-casing?
4. **Both quasi-static limits in tandem** (§7): electroquasistatic from the  $\Phi$  equation, magnetoquasistatic from the **A** equation, both accurate at once.
5. **Factorization cost and memory** at case 05/06 scale.
6. **A genus > 0 case** — case 05's ring — for the null magnetostatic solution.
7. **Solver-independence**, following Ansari (§3.1): solve the same gauged system with a direct and an iterative solver and require the *potentials* to agree, not just the fields. It is a cheap check and it is the one that exposed the problem for him.

**And one measurement to take first, before any gauge work begins.** Build a minimal case with a port face **inside** the domain crossing conductor and dielectric, and run case 07's tree-permutation harness on it. Two outcomes, both worth having:

- **$\Phi$  and the terminal quantities move with the tree** — the predicted failure is

now *measured* in this project rather than inherited from the literature, and there is a quantitative baseline the new gauge must fix.

- **They do not move** — then the premise behind the whole gauge programme needs

re-examining before anything is rewritten.

This is cheap: the harness exists, the solver exists, and the case is smaller than case 06. It should run at low frequency, where §1.2 argues the failure already appears.

---

---

## 10. Corrections to the project's own bibliography

Four entries in `Gemini/APhi_FEM_BEM_Comprehensive_Report_Updated.tex` are wrong and have propagated into the proposal's novelty claims:

| CLAIM                                                                                     | CORRECTION                                                                                                                                                                |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Bogaert, Cools & Andriulli, arXiv:1705.00265"                                            | That arXiv ID is <b>Li, Fu &amp; Shanker</b> , <i>Potential Integral Equations in Electromagnetics</i> .                                                                  |
| Abdrabou & Gomez use "structured dual grids"                                              | They use <b>unstructured meshes</b> . The real distinction is <b>0-form Cartesian components vs Whitney 1-forms</b> — the novelty claim must be re-grounded on that axis. |
| The generalized Lorenz gauge is what reduces the boundary system to <b>V</b> and <b>K</b> | The <b>Cartesian scalar reformulation</b> performs the reduction; the gauge enables it. Two steps.                                                                        |
| Chew 2014: "Generalized Lorenz Gauge ... Anisotropic Media", pp. 69–77                    | Actual title <i>Vector Potential Electromagnetics with Generalized Gauge for Inhomogeneous Media: Formulation</i> , pp. 69–84.                                            |

Also: `AphiFreqDomainTreeCotree_SeungJFlee2003.pdf` is mis-filed — it is Lee, Lee & Lee, *Hierarchical Vector Finite Elements for Analyzing Waveguiding Structures*, IEEE T-MTT **51**(8) 2003, a 2-D waveguide eigenanalysis paper that says nothing about  $\Phi$ 's non-uniqueness.

---

## 11. The Darmstadt / ABB / Siemens line — the literature does settle the spike's finding 2

Added after `GENERALIZED_LORENZ_GAUGE.md` §11 reported that  $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$  wrecks conditioning below ~1 GHz and concluded "**this is now the main obstacle and nothing should be implemented before it is settled.**" That conclusion was reached without checking the roadmap's own reference list, and it is wrong in its premise: the phenomenon is named, published, and has remedies. It is a *known* obstacle, not a new one.

Four papers, one continuous research line (TU Darmstadt + ABB + Siemens Digital Industries), none of which appeared in §8's ranking.

## 11.1 Balian et al. (2023) — our finding 2, by name, with a remedy

*"When simulating resistive-capacitive circuits or electro-quasistatic problems **where conductors and insulators coexist**, one observes that large time steps or **low frequencies lead to numerical instabilities, which are related to the condition number of the system matrix**. Here, we propose several stable formulations **by scaling the equation systems**."*

That is the spike's finding 2 restated as a paper's abstract, and the mechanism they name is the same one: the  $\sigma/\omega\epsilon$  ratio between conducting and insulating regions.

**The remedy is a scaling applied analytically, before assembly.** Multiply equation block  $k$  by  $a_k$  and substitute the unknown  $\psi_k = b_k^{-1} \phi_k$ :

|                                  |                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------|
| (i) symmetric                    | $a_2 = b_2 = \omega^{(-1/2)}$ , $a_1 = b_1 = 1$                                                     |
| (ii) non-symmetric               | $a_2 = \omega^{-1}$ , rest 1                                                                        |
| (iii) symmetric with material    | $a_1 = b_1 = (\sigma_1 + j\omega\epsilon_1)^{(-1/2)}$ , $a_2 = b_2 = (\epsilon_2 j\omega)^{(-1/2)}$ |
| (iv) non-symmetric with material | $a_1 = (\sigma_1 + j\omega\epsilon_1)^{-1}$ , $a_2 = (\epsilon_2 j\omega)^{-1}$ , $b = 1$           |

Three things in that table matter to us:

1.  **$a_k = b_k$**  is a congruence — it scales the equation *and* the unknown, so symmetry survives. The implementation spec's §11 framed our choice as "symmetric or well-conditioned, not both". Variants (i) and (iii) are both. §11.4 below is the qualification on how far that carries to our matrices.
2. **Variant (iii) scales by  $(\sigma + j\omega\epsilon)^{(-1/2)}$**  — exactly our  $\beta$ . The material combination the spike found fatal is the one they scale *by*.
3. **It must be done before assembly**, symbolically: *"the products of powers of  $\omega$  must be determined before matrix assembly to avoid numerical errors or division by zero."* A post-assembly Jacobi preconditioner is variant (iv) applied too late (their §III, citing [17, §4.1]).

Measured: the unscaled RC system breaks down below 1e10 Hz; (i) and (iii) stay flat to 0 Hz, with (iii) better than (i) by  $1/(2RC) \approx 5e11$ . Two caveats the authors state themselves — **(i) destabilizes above 1e10 Hz**, where the problem becomes essentially capacitive, and under (i)/(iii)  **$\phi$**  cannot be recovered at exactly  $\omega = 0$  because the unknown is the scaled one, *"a natural consequence of the fact that it is not well-defined from the start."* Variants (ii)/(iv) keep the original unknowns and lose symmetry.

This covers the  $\Phi$  equation only.

## 11.2 Herles et al. (2025) – the A equation, and it keeps tree–cotree

arXiv:2502.13588, IEEE Trans. Magn. The companion that stabilizes the other half; they say so outright — *"Effective modifications are proposed in [14] to improve the condition number. The focus of our paper is the stabilization of (11)",* (11) being the curl–curl system.

Their scheme is Ostrowski & Hiptmair's two-step (§11.3): solve an electro-quasistatic problem for  $\phi$ , then use it as the source for  $A$ , with **the EQS problem itself serving as the gauge condition**. Taking  $\text{div}$  of the  $A$  equation and dividing by  $i\omega$  gives the implicit constraint

$$\text{div}(\kappa A) = 0, \quad \kappa = \sigma + i\omega\epsilon \quad (17)$$

a **generalized Coulomb gauge with complex conductivity**. It degenerates in the insulator as  $\omega \rightarrow 0$ , so they split it by region:

$$\begin{aligned} \alpha \text{div}((\sigma + i\omega\epsilon) A) &= 0 & \text{in } \Omega_C & \text{(conductor)} \\ \beta \text{div}(\epsilon A) &= 0 & \text{in } \Omega_A & \text{(insulator)} \end{aligned} \quad (18)$$

**Each piece is frequency-independent in its own region.** This is the structural option the spike never considered: *a different gauge condition inside conductors than outside*. Chew carries one  $\chi$  everywhere, and the spec's §11 listed "a different  $\chi$  inside conductors" as a speculative line item — here it is, built and measured. Their weights are  $\alpha = 1 + \omega$  and  $\beta = (1 + \omega) \max \sigma + \sigma_{\text{art}} \max \epsilon$  with  $\sigma_{\text{art}} = 1e-6$ : **the scaling carries the peak material magnitude**, the same idea as Balian's variant (iii).

The discretization is the part worth copying. Build the weighted divergence matrix  $(S_?)_{ij} = \int_? \text{div}(w_j) v_i dV$ , take the tree–cotree split of the edge DOFs, then **replace the redundant tree rows of the curl–curl system with the gauge-constraint rows**:

$$\begin{bmatrix} W^{(RR)} & W^{(RT)} \\ S^{(R)} & S^{(T)} \end{bmatrix} \begin{bmatrix} a^{(R)} \\ a^{(T)} \end{bmatrix} = \begin{bmatrix} j^{(R)}(u) \\ 0 \end{bmatrix} \quad (26)$$

They prove  $W^{(RR)}$  has full rank even at  $\omega = 0$ , and the second row of the cotree system is automatically satisfied, so nothing is lost by the swap. The symmetric alternative is the Lagrange-multiplier saddle point (21), which they reject on size.

**Tree–cotree is not discarded — it is what tells them which rows are redundant.** That is a very different and much cheaper change than replacing the gauge: we already have a spanning tree.

Measured, on a mesh of **three conducting bars in a dielectric box** with  $\phi = 0$  and  $\phi = 1$  on opposite faces:

| COND AT F = 0 |          |
|---------------|----------|
| original      | singular |
| stabilized    | 4.87e5   |

and on a copper planar coil ( $\sigma = 6e7$ ),  $1.9e9$  against singular. The discrete gauge residual  $\|S \cdot a\|_2$  stays below  $1e-11$  across the whole sweep and is  $1.52e-12$  at DC, against  $\sim 1e-2$  unstabilized. Fig. 8 is the one to look at: the unstabilized  $\|B\|$  and  $\|E\|$  are **plotted logarithmically because they reach  $1e70$  and  $1e75$**  — garbage of exactly the kind a mixed-material port produces.

Two further details bear on our problem directly:

- **Their DOF partition assigns interface nodes to the conductor.** Index sets

(12)–(13):  $I_v^*(A)$  is the nodes whose support does *not* intersect the conductor, and  $I_v^*(C)$  is *everything else*, so a node on the conductor/dielectric interface is a conductor node. They do not try to make a per-element material factor behave — they partition, then scale blockwise.

- **Their §V-D is close to our case 05.** A thin dielectric slit inside a

conducting loop ( $\epsilon_r = 7.2e15$ , acting as a lumped capacitor), with the current closing through the gap via  $i\omega D_m$ . A dielectric gap interior to a conductor, carrying current — structurally our internal cut — handled by the generalized gauge.

### 11.3 Ostrowski & Hiptmair (2020/2021) — the origin

ETH SAM Research Report 2020-43, published as SIAM J. Sci. Comput. **43**(4) B1008–B1028 (2021). Introduces EQS-as-gauge-condition and the two-step procedure, with frequency-stable weak forms for both steps:

*"the electro-quasistatic fields can be corrected for magnetic/inductive phenomena at any frequency in a second step. The combined field from both steps is a solution of the full Maxwell's equations... Electro-quasistatics serves as a gauge condition in this semi-decoupled procedure."*

Structurally this is Chew's move — an independent  $\Phi$  equation, then an  $A$  equation taking  $\Phi$  as a source — reached from the low-frequency side with a different gauge condition and no  $\chi$ .

## 11.4 What transfers to Chew's formulation, and what does not

Being careful here, because these papers stabilize the **generalized Coulomb** gauge, not Chew's generalized Lorenz, and §8's ranking must not be rewritten on a false equivalence.

### Transfers directly:

|                              |                                                                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the diagnosis                | $\sigma/\omega\epsilon$ contrast as the mechanism is confirmed and published — not a bug in our spike. Herles §III-B states it in a line: <i>"The ratio <math>\sigma/\omega\epsilon \gg 1</math> in <math>\Omega_C</math> increases this issue even further."</i> |
| the technique                | analytic pre-assembly scaling by material-dependent powers of $\omega$ is formulation-independent, and applies to the §7 block system as written.                                                                                                                 |
| $a_k = b_k$                  | a two-sided congruence keeps symmetry. The spike's dichotomy was too quick.                                                                                                                                                                                       |
| per-region gauge             | a different gauge condition in conductor and insulator is a real, measured option, not speculation.                                                                                                                                                               |
| partition, don't diagonalize | assign interface DOFs to the conductor set and scale blockwise.                                                                                                                                                                                                   |

### Does not transfer, and this is the honest limit:

- Our symmetry problem is **not** a block scaling.

$K_{NE}[m,n] = -\int (\epsilon/\chi) \nabla \lambda_m \cdot \omega_n$  against  $K_{EN}[m,n] = \int \epsilon \omega_m \cdot \nabla \lambda_n$ : the factor  $\epsilon/\chi$  sits *inside the element integral*. If it is constant per region, then for a node strictly interior to one region every element in its support shares the factor, and that row of  $K_{NE}$  is exactly  $-c$  times the matching row of  $K_{EN}^T$ . **For a node on a material interface it is not**, under

any diagonal scaling. So a congruence repairs symmetry everywhere except at interface nodes — which is precisely where a mixed port lives. Balian and Herles sidestep this by assigning those nodes to one block, not by exactness.

- **Neither paper addresses a port on a mixed-material face.** Herles's dielectric

slit is an interior gap with no potential boundary condition on it. §11.2's evidence is suggestive for case 05, not an answer for the coax via.

- Their target is DC-to-MHz industrial devices. \*\*Nothing here is validated at

THz\*\*, which is the stated end goal.

### 11.5 What this changes

1. \*\* `GENERALIZED_LORENZ_GAUGE.md` §11 is corrected\*\* — the  $\sigma$  question is a known problem with published remedies, not a blocker on implementation.
2. **A cheaper route exists and belongs in §8's ranking:** keep our spanning tree, add the weighted divergence matrix  $S$ , swap the tree rows per Herles (26). That is incremental on the solver we have, where Chew is a rewrite. It gives up symmetry as (26) is written, and it is a Coulomb-family gauge, so it inherits the  $\hat{n} \cdot A$  continuity problem of §6 — the reason we went looking in the first place. **It is not a substitute for the mixed-port fix; it is a substitute for the low-frequency-conditioning half of the argument**, and those two motivations should stop being bundled.
3. **Reference code exists:** Herles, *Low Frequency Stable Full Maxwell*, doi:10.5281/zenodo.14810885 — GeoPDEs / Octave. Worth reading before writing our own  $S$ .

## 11.6 Still not obtained

### WHY IT MATTERS

|                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eller, Reitzinger, Schöps &amp; Zaglmayr (2017)</b> , SIAM J. Sci. Comput. <b>39</b> (4) B703–B731, doi:10.1137/16M1077817                                                              | <i>"monolithic, symmetric, low-frequency stable, broadband... no auxiliary variables... stable even if the frequency equals zero."</i> Symmetric <b>and</b> stable <b>and</b> monolithic answers all three of the spike's complaints at once. Paywalled, no preprint — but <b>not worth buying</b> , because §11.8 obtained its method from two open sources. |
| <b>Jochum, Farle &amp; Dyczij-Edlinger (2015)</b> , IEEE Trans. Magn. <b>51</b> (3) 7402304, and the 2016 SCEE companion <i>A symmetric and low-frequency stable potential formulation</i> | Previously flagged, still unread.                                                                                                                                                                                                                                                                                                                             |
| <b>Demerdash &amp; Wang (1990)</b>                                                                                                                                                         | Coulomb-gauge breakdown at permeability contrast. Still unread.                                                                                                                                                                                                                                                                                               |
| <b>Zhu &amp; Jiao (2010)</b> , IEEE Trans. Adv. Packag. <b>33</b> (4) 1043–1050                                                                                                            | <i>"theoretically rigorous full-wave FEM solution of Maxwell's equations from dc to high frequencies"</i> — the reference both Darmstadt papers cite for the breakdown itself.                                                                                                                                                                                |
| <b>Manges &amp; Cendes (1995)</b> , IEEE Trans. Magn. <b>31</b> (3) 1342–1347                                                                                                              | <i>A generalized tree-cotree gauge for magnetic field computation</i> — origin of the term Herles et al. use.                                                                                                                                                                                                                                                 |
| <b>Hiptmair, Kramer &amp; Ostrowski (2008)</b> , IEEE Trans. Magn. <b>44</b> (6) 682–685                                                                                                   | <i>A robust Maxwell formulation for all frequencies.</i>                                                                                                                                                                                                                                                                                                      |
| <b>Chew's [28] White &amp; Koning (2002) and [29] Dai, Chew &amp; Jiang (2013)</b>                                                                                                         | Sources for the $\bar{\mathbf{G}}^{-1}$ elimination. Still unread.                                                                                                                                                                                                                                                                                            |

## 11.7 New in APhi\_Papers/

### FILE

|                                                         |                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Balian2023_LowFreqStab_ConductorsInsulators.pdf         | Balian, Merkel, Ostrowski, De Gersem & Schöps, arXiv:2302.00313, IEEE Trans. Dielectr. Electr. Insul. <b>30</b> (6), 2023. §11.1. **Downloaded under DyczijEdlinger2023_... and renamed** — the roadmap attributes it to Dyczij-Edlinger, who is not an author. |
| TwoStep_GeneralizedTreeCotree_LowFreqStability_2025.pdf | Herles, Mally, Ostrowski, Schöps & Merkel, arXiv:2502.13588. §11.2.                                                                                                                                                                                             |
| ETH2020_FreqStableFullMaxwell_EQSGauge.pdf              | Ostrowski & Hiptmair, ETH SAM 2020-43. §11.3.                                                                                                                                                                                                                   |
| TwoStep_TimeDomain_Stabilized_2025.pdf                  | arXiv:2507.18235, same group, time-domain extension. Not relevant to a frequency-domain solver; filed for completeness.                                                                                                                                         |
| Clemens2022_DarwinTypeQuasistatic.pdf                   | arXiv:2204.06286, Darwin-type quasistatic formulations. Cited by both; unread.                                                                                                                                                                                  |
| ShinFan2013_EigenvalueEngineering_0E21.pdf              | Source of Chew's $\alpha > 0$ eigenvalue claim. Extracted, unread.                                                                                                                                                                                              |

## 11.8 Eller's method, obtained without Eller — and a competing design for our exact application

Both sources below are open access and were obtained after §11.6 listed Eller et al. as the paper we most wanted. **Eller no longer needs to be bought.**

Stysch2022\_BroadbandFEM\_Impedance\_ParasiticExtraction.pdf

Stysch, Klaedtke & De Gersem, *Electrical Engineering* **104**(2) 855–867, 2022, doi:10.1007/s00202-021-01348-9. Also arXiv:2009.08232 — it was free all along.

Stysch2022\_PhDThesis\_StableBroadbandParasiticExtraction.pdf

Stysch, PhD thesis, TU Darmstadt, 2022, handle [tuda/8889](#).

**Chapter 6, 24 pages, is a full treatment of Eller's scheme,** including §6.6.1 tree–cotree splitting, §6.6.2 gradient-space splitting, and §6.3.2 *Stabilizing the Lorenz-gauged system matrix* — which Eller's own paper, being E-field only, does not cover.

Robert Bosch GmbH + TU Darmstadt (De Gersem is also a co-author of Balian et al., §11.1). The application is **parasitic extraction for EMC** — ours.

### Eller's mechanism, as the article states it

Split the trial and test space three ways,  $H(\text{curl}, \Omega) = V \oplus W \oplus U$ , with

$$V : \int_{\Omega} |\text{curl } v|^2 dV \neq 0 \quad (36a)$$

$$W : \text{curl } w = 0 \quad \text{and} \quad \int_{\Omega_c} |w|^2 dV \neq 0 \quad (36b)$$

$$U := \{ u \in H(\text{curl}, \Omega) : \text{curl } u = 0 \wedge u = 0 \text{ in } \Omega_c \} \quad (36c)$$

so  $V$  is the part with real curl,  $W$  the gradient fields that live in the conductor,  $U$  the gradient fields that vanish there. Then scale the three parts by **different powers of  $\omega$** :

$$E = j\omega E_V + (j\omega)^{(1/2)} E_W + E_U \quad (37)$$

and test the equation separately with each subspace, which scales each of the three resulting equations independently. \*\*That fractional  $\omega^{(1/2)}$  is what Balian et al. meant\*\* by “a scaling by fractional powers of  $\omega$  was similarly applied in [12]” — their [12] is Eller et al. Same family as §11.1's variant (i), applied to a three-way space split rather than a two-way material one.

**\*\*At lowest order the  $V$  basis is found by a tree–cotree split\*\*** (their [19]). That is now the third independent paper in this section that keeps the spanning tree and changes only what is done with it.

### The part that bears on case 06 and 07 directly

§2 of the article is the cleanest published statement of what our gauge tests measured. They start from the problem by name — *"To reconcile the path-dependent voltage concept of electromagnetic field theory with the path-independent voltage concept of electrical circuits poses a challenge"* — and resolve it by subtracting the return path's partial inductance:

$$V_c := V - j\omega I L_r = \int_c E \cdot dl + j\omega \int_c A \cdot dl = \Phi(T_b) - \Phi(T_a)$$

their (5)–(7). So the terminal  $\Phi$ -difference **is** a path-independent quantity, and it is *not* the full terminal voltage: it is the voltage minus the return path's partial inductance. And then, explicitly:

*"The values of these partial inductances depend on the gauge condition chosen for the magnetic vector potential  $A$  and electric scalar potential  $\Phi$ ." "As it incorporates a partial inductance,  $V_c$  formally depends on the gauge condition of potentials  $\Phi$  and  $A$ ."*

**This is our  $\omega L/R$  mechanism, published.** The gauge-sensitive part of a terminal  $\Phi$ -difference is exactly the split of the loop inductance into the conductor's partial inductance and the return path's —  $L_{\text{loop}} = L_c + L_r$  is gauge-invariant while the two terms separately are not (their (4)). **07\_GaugeInvariance** measured the same thing from the other end.

The consequence for how we present our results: a tree-dependent  $\Phi$  is **not simply an error to be driven to zero**. For parasitic extraction the partial inductance is the quantity you want, and the gauge is what *defines* how the loop inductance is attributed between conductor and return path. It is a modelling convention that must be fixed, not a defect. What is a defect is leaving it *implicitly* fixed by a spanning-tree traversal order, which is what we do today.

**How they make  $\Phi$  unique — the answer to the question that started this file**

They impose the gauge as **an explicit PDE for  $\Phi$** , and solve it. Lorenz gauge (their (12),  $\text{div}(\mathbf{v}_r \mathbf{A}) + j\omega\Phi/c^2 = 0$ ) combined with  $\mathbf{E} = -\text{grad } \Phi - j\omega\mathbf{A}$  eliminates  $\mathbf{A}$  and leaves a scalar boundary-value problem:

$$-\text{div}(\mathbf{v}_r \text{ grad } \Phi) - (\omega^2/c^2) \Phi = \text{div}(\mathbf{v}_r \mathbf{E}) \quad (13)$$

with, after the inductive compensation term is added (their (27)):

$$\begin{aligned} \Phi_c &= \text{const.} && \text{on } \Gamma_{el} \\ \hat{n} \cdot \mathbf{v}_r \text{ grad } \Phi_c &= 0 && \text{on } \Gamma_{mag} \end{aligned}$$

**$\Phi = \text{const.}$**  on the electrode, with the constant not prescribed, is a floating potential condition\*\* — the "single potential port, no ground defined" behaviour you described from Maxwell, as a boundary condition rather than a special port type.

So  $\Phi$  is unique because it solves its own well-posed BVP. No tree, no gauge ambiguity in  $\Phi$  at all. The gauge condition has become the equation that determines it. Structurally this is Chew's decoupled  $\Phi$ -equation and Ostrowski & Hiptmair's EQS-as-gauge (§11.3) reached a third way.

Their full procedure is three sequential solves —  $\mathbf{g}$ , then  $\mathbf{E}$ , then  $\Phi_c$  (their Fig. 2) — and the MQS limit is reached by dropping the  $\omega^2$  term, which they note *"is equivalent to choosing the Coulomb gauge condition for the calculation of  $\Phi$ "*, necessary at high frequency for consistency between the two PDEs.

### Why this is a competing design and not just a reference

They considered our formulation and rejected it. Quoting their §3.1 on the two options for computing  $\Phi$ :

*"It is more advantageous to use the latter 'E approach' for several reasons: The two fields  $E$  and  $\Phi$  can be calculated in sequence... thereby avoiding a computationally more expensive coupled boundary value problem (BVP), which occurs in the 'A- $\Phi$  approach'. Furthermore, the E approach allows for an easy treatment of conductors modeled as perfect electric conductors... Finally, the use of the E-field formulation allows for an elegant stabilization of the low-frequency instability."*

Three reasons, from the same institute, for the same application. This belongs in §8's ranking as a genuine alternative to  $A-\Phi$ , not a footnote. Against it, for our purposes:  $E$  in  $H(\text{curl})$  gives up the potentials we need for a potential-based BI coupling (§4), and their own §4.1.1 shows the volumetric discretization saturating the resistance above ~10 MHz, which is a hard limit for a THz target.

### Still unread here

Chapter 6 of the thesis. Neither our extractor nor TUpriints' own text layer recovers it — TUpriints' stops mid-chapter 5, and the thesis PDF yields only 14 k characters from 148 streams because of its font encoding. The PDF renders normally and can be read page by page; chapter 6 spans pp. 63–86.

---

## 11.9 Where this leaves the decision

Three papers now keep tree–cotree and change what is done with it (Herles, Eller/Stysch, and the Manges–Cendes line they both cite), and two independent groups obtain a unique  $\Phi$  by **making the gauge condition an explicit equation for  $\Phi$**  rather than by fixing degrees of freedom on a tree (Chew via  $\chi$ , Stysch via the Lorenz PDE, Ostrowski & Hiptmair via EQS). That convergence is the strongest signal in this document, and it points at the same place from three directions: **stop setting tree DOFs arbitrarily and give  $\Phi$  its own equation.**

What that does *not* settle is the mixed-material port, which is still the project's actual blocker. Stysch's  $\Phi_c = \text{const.}$  on an electrode is the closest published thing to the floating-potential port, and it is a boundary condition on a scalar BVP — which is only available once  $\Phi$  has its own equation. The two problems may therefore have one solution, but that is a conjecture and is not demonstrated by anything read so far.

---

## 11.10 Chapter 6 of the thesis, read in full — and it contains the floating port

§11.8 left chapter 6 unread because the thesis PDF defeated our extractor. It is a CID-font document: text is hex-coded glyph ids, not characters, and `ex2.py` only understands literal strings, so it recovered 14 k characters from a 155-page thesis.

`tools/pdftext_cid.py` decodes through each font's `/ToUnicode` CMap (306 of them here) and splits output by page: 267 k characters, and chapter 6 is pp. 63–86, which are PDF pages 75–98.

Two sections of it are **directly implementable in our solver today, whichever gauge we end up with.**

### 11.10.1 What Eller's method actually is

Not a new formulation — *a synthesis of two existing ones* (thesis p. 64):

| FROM                                                 | WHAT IT CONTRIBUTES                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hiptmair, Kramer &amp; Ostrowski (2008)</b>       | the splitting of the scalar space <code>H<sup>1</sup></code> . Their own "generating systems" approach enforces Gauss's law in the non-conducting domain only and the Coulomb gauge in the conductor — but yields matrices that are <b>singular and non-symmetric</b> .                                                                                                                                                                          |
| <b>Jochum, Farle &amp; Dyczij-Edlinger (2015/16)</b> | the view that <i>**tree–cotree</i> is a Helmholtz-type decomposition of <code>H(curl)</code> **. Their formulation is symmetric and regular even in the lossy case, by treating conducting and non-conducting regions with separate equations plus interface conditions on <code>∂Ω<sub>c</sub></code> — at the cost of extra DoFs in conductors and <i>"significantly less sparse matrices, increasing the computation times considerably."</i> |

Eller unites the two splittings. Note what this does to §11.6's wish-list: the two Jochum papers we still have not read are *inside* this, and **in the PEC case Eller's equations and Jochum's are equivalent** (p. 64).

The decomposition is two-stage. First a Helmholtz split off the first level of the de Rham complex, then a split of the gradient part by whether it survives in the conductor:

$$\begin{aligned}
 H(\text{curl}, \Omega)' &= V \oplus Y & (6.11) \\
 V : \quad &\langle \text{curl } v, \text{curl } v \rangle \neq 0 & \text{(functions with real curl)} \\
 Y : \quad &\text{curl } w = 0 & \text{(gradient fields)}
 \end{aligned}$$

$$\begin{aligned}
 Y &= W \oplus U & (6.12) \\
 W : \quad &\langle w, w \rangle_{\Omega_c} \neq 0 & \text{(gradients that live in the conductor)} \\
 U : \quad &u = 0 \text{ in } \Omega_c & \text{(gradients that vanish there)}
 \end{aligned}$$

$$H(\text{curl}, \Omega)' = V \oplus W \oplus U \quad (6.13)$$

Then scale the three parts by different powers of the wave number `k = ω/c` and test each with its own subspace:

$$\mathbf{E} = j\mu_0 c \left( \mathbf{k} E_V + \nabla E_W + E_U \right) \quad (6.14)$$

$j$  is folded into the scaling deliberately, so that the discretized system becomes **real** in the PEC case.

The sentence that matters most to us (p. 67):

"The magnetic flux density can be recovered with the simple expression  $\mathbf{B} = -\mu_0 \text{curl } E_V$ , illustrating that  $E_V$  is essentially a MVP\*\*, while  $E_W$  and  $E_U$  can be understood as **gradient fields produced by ESPs**."

So Eller's three-way split of  $\mathbf{E}$  is an  $A-\Phi$  decomposition in disguise —  $E_V$  plays  $A$ , and  $E_W / E_U$  are  $\nabla\Phi$  separated by whether the potential survives inside the conductor. The apparent fork between "the E-field route" and "our  $A-\Phi$  route" is much narrower than §11.8 implied: the space structure is the same one, and the scaling (6.14) is a statement about the **relative weighting of  $A$  against  $\nabla\Phi$** , which is a thing we can apply directly.

The resulting weak system (6.15a–c) is **symmetric**. It is LF-stable in the sense that the matrix has full rank at every frequency — but the right-hand sides of (6.15b,c) still diverge as  $k \rightarrow 0$ , so a second stabilization solves the static fields  $F_U$  (electrostatic) and  $F_W$  (static current) first, in the static limit, and writes  $\mathbf{E}$  as those plus non-static corrections (6.17). That gives (6.22), with the same operator and a bounded RHS. A useful special case: \*\*if both terminals lie on the same conductor,  $F_U = 0$  \*\* and the impedance can be evaluated exactly at DC.

### 11.10.2 Our $\epsilon_{\text{eff}}$ instability, named, in their equation (6.1)

The spike's finding 2 appears here as a one-line diagnosis. Enforcing Gauss's law explicitly over all of  $\Omega$  in a potential formulation gives

$$\text{div} [ (\epsilon + \sigma/j\omega)(\text{grad } \phi + j\omega\mathbf{A}) ] = (1/j\omega) \text{div } \mathbf{J}_s \quad (6.1)$$

and the thesis says of it: "This equation contains the unstable term  $(j\omega)^{-1} \text{div } \sigma \text{ grad } \phi$  on its left-hand side." Plus, for a source current that is not divergence-free, the RHS diverges too.

$\epsilon + \sigma/j\omega$  is our  $\epsilon_{\text{eff}}$ . \*\*The unstable object is specifically  $(j\omega)^{-1} \text{div}(\sigma \nabla \phi)$  \*\* — not the material contrast in the abstract, but that one term. That is a sharper statement than `GENERALIZED_LORENZ_GAUGE.md` §11 reached by measurement, and it says where to look: the  $\sigma$ -weighted gradient block of the  $\Phi$  equation, at the conductor boundary.

11.10.3 Measured

Conductor segment, 10 mm × 1 mm², 680 elements, so the condition number and the *rank deficiency* are both computed exactly:

|                           | UNSTABLE (6.8)                                                 | STABILIZED (6.22)                                |
|---------------------------|----------------------------------------------------------------|--------------------------------------------------|
| matrix dimension          | 4518                                                           | 4519 — <b>one extra DoF</b>                      |
| condition number          | rises sharply below 10 kHz, then plateaus on machine precision | approximately <b>constant at all frequencies</b> |
| rank deficiency at 10 kHz | 0                                                              | 0                                                |
| rank deficiency at 1 kHz  | <b>630</b>                                                     | 0                                                |

On a capacitor–coil model the unstabilized impedance is "*strongly scattered*" below ~100 kHz; at 5 kHz the unstabilized  $|E|$  runs to 1e8–1e15 V/m with an implausible distribution (their Fig. 6.6).

Two results worth more than the stabilization itself:

- \*\*Fig. 6.7: at high frequency the stabilized and conventional systems give identical values\*\*, so "*no switching of systems is necessary.*" One formulation covers DC to resonance. That is the property our roadmap wants and that Balian's variant (i) explicitly does *not* have (§11.1: it destabilizes above 1e10 Hz).

- **Cost, their Table 6.1**, on a 520 k-DoF wire model:

| SYSTEM    | N (STAB/UNSTAB) | NON-ZEROS       | TIME            | RATIO        |
|-----------|-----------------|-----------------|-----------------|--------------|
| full wave | 520 k / 520 k   | 27.3 M / 22.7 M | 86.5 s / 67.9 s | <b>1.27×</b> |
| MQS       | 435 k / 520 k   | 13.7 M / 13.9 M | 36.1 s / 43.0 s | <b>0.84×</b> |
| MS (PEC)  | 336 k / 520 k   | 8.71 M / 9.54 M | 38.2 s / 44.7 s | <b>0.86×</b> |

Stabilization costs 27 % in the full-wave system and **saves** 14–16 % in the quasistatic ones. The +20 % non-zeros in the full-wave row are attributed specifically to constructing  $U_h$  — §11.10.5 below, which is the part that densifies the matrix.

#### 11.10.4 The tree-construction rules Eller omitted

§6.6.1, citing Klis 2015, and flagged in the thesis as *"crucial details especially with regards to the tree creation that were omitted in [Ell+17; Ell17]"* — i.e. **the thesis is strictly better than the paper for implementing this**, which settles §11.6's question about buying Eller.

The rules, for a boundary split into magnetic, electric and absorbing parts:

|                           |                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Gamma_{\text{mag}}$     | needs no special treatment                                                                                                                                                                         |
| $\Gamma_{\text{el}}$      | edge DoFs on it are set to zero                                                                                                                                                                    |
| $\Gamma_a$<br>(absorbing) | its basis functions <b>must</b> belong to $V_h$ , by (6.11b)                                                                                                                                       |
| consequence               | <b>**edges of <math>\Gamma_{\text{el}}</math> and <math>\Gamma_a</math> must be assigned to the tree**</b>                                                                                         |
| compatibility             | for each <i>disconnected</i> part $\Gamma_i$ of $\Gamma_{\text{el}} \cup \Gamma_a$ , <b>**contract all of its mesh nodes to a single vertex <math>v_i</math> in the graph**</b> before spanning it |

In the PEC case each conductor's surface is one such disconnected boundary. Then the ordering rule, which is the kind of thing that only shows up when someone has actually built it:

**"numerical artifacts can arise if in the non-conducting domain  $\Omega_o$  the tree would not be equivalent to a PEC case tree that contains only one vertex  $v_i$  per disconnected boundary. The problem is avoided by **\*\*first constructing a tree in  $\Omega_o$  respecting the boundaries of (6.50) and subsequently adding the edges of trees constructed in the individual conductors\*\***."**

So: contract each boundary component to one vertex, span the **insulator first**, then span inside each conductor. Our tree is built by a traversal over the whole mesh with no such structure, and **07\_GaugeInvariance** measures the consequence. **This is implementable now and is independent of the gauge decision.**

#### 11.10.5 The floating-potential port, as a congruence

§6.6.2. This is the answer to the port question that has been open since you described Maxwell's double-potential floating ports, and it arrives from an unexpected direction — it is not a port model at all, it is how they *construct the space*  $U_h$  of gradient fields that vanish in the conductors.

A potential  $\psi_u$  whose gradient vanishes in  $\Omega_c$  must be **constant on each disconnected conductor**  $\Omega_{c,i}$ , and on each disconnected part of  $\Gamma_{e1}$  and  $\Gamma_a$ . Call the number of such entities  $n_y$ . Then:

**\*"The first constant can be chosen zero, and all  $n_y - 1$  others **must become DoFs in the linear system.**"\***

One scalar unknown per floating entity, with the first grounded. That *is* a floating-potential port. The implementation (6.51):

1. discretize with the ordinary nodal gradient basis, giving  $\hat{A}$ ,  $\hat{b}$ ;
2. zero the DoFs of the first entity, and zero all **higher-order** DoFs on the remaining  $n_y - 1$  entities, since first order already suffices to represent a constant;
3. aggregate the first-order DoFs of each remaining entity onto one DoF with a 0/1 matrix  $P$  (their example (6.52) is a row of ones over one conductor's nodes), and form

$$A = P \hat{A} P^T, \quad b = P \hat{b}, \quad \text{recover } \hat{x} = P^T x \quad (6.51)$$

**\*\*  $A = P \hat{A} P^T$  is a congruence, so symmetry is preserved exactly\*\*** — and  $P$  is *not diagonal*. That is the general form of the device §11.4 was looking for when it concluded a diagonal congruence fails at material-interface nodes: the right object is an **aggregation** matrix, not a scaling. It costs sparsity (the +20 % non-zeros of §11.10.3) because each aggregated row is dense over that conductor's nodes.

Three things follow:

- **A floating port needs no new formulation and no new gauge.** It is one

aggregated nodal DoF per floating conductor, applied as a congruence. This can be built on the solver we have.

- It answers your question **\*"do we work on floating double potential ports first,**

then the gauge, or both together?"\* — **they are separable**, and the port is the smaller, lower-risk piece.

- It does **not** by itself fix the mixed-material port. Aggregating a face's

nodes to one DoF presumes that face is an equipotential, which is what a conductor electrode is and a mixed conductor/dielectric face is not. For the coax via the dielectric annulus must stay free ( `COAX_PORT_ANALYSIS.md` line 66 already says so). So this gives us the floating *conductor* port, and leaves the mixed port where it was.

### 11.10.6 And the $\Phi$ equation is the easy half

§6.3 states plainly that the BVP for  $\Phi$  "*does not possess a general LF breakdown*": if the boundary is not purely magnetic the matrix has full rank at all frequencies, and if it is, only the spatially constant component of  $\Phi$  is undetermined — fixed by pinning  $\Phi$  to zero at one arbitrary point (their (6.29)).

The one exception is the **Lorenz-gauged**  $\Phi$  equation with a purely magnetic boundary, where the  $-k^2 \langle \Phi, \psi \rangle$  mass term fixes the constant at finite frequency but vanishes as  $k \rightarrow 0$ , leaving a rank deficiency of exactly 1 — harmless for an iterative solver, fatal for a direct one. Their fix (§6.3.2) is the same move again, one dimension smaller:

$$H^1(\Omega) = H^1_f(\Omega) \oplus \mathbb{C} \quad (6.30)$$

$$\Phi_c = j\mu_0 c \left( \Phi_v + \Phi_o / k \right) \quad (6.31)$$

— split off the constant mode as **its own unknown, scaled by  $1/k$** , and test with all of  $H^1_f$  plus the single constant function **1**. The result (6.32) is "*still symmetric but less sparse*". Same pattern as §11.10.5 and as Balian's variant (i): isolate the mode that degenerates, give it its own row, and scale it by the power of  $\omega$  that keeps it finite.

### 11.10.7 New references from this chapter

|                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Albanese &amp; Rubinacci (1988)</b>                                                                                                                                                                      | "a seminal paper" — the origin of tree–cotree, earlier than the Manges & Cendes 1995 generalization in §11.6.                                                                                                                                                                                                                                  |
| <b>Klis (2015)</b> , <i>Schnelle Finite-Elemente-Verfahren für verschiedene Klassen magneto-quasistatischer Probleme</i> , dissertation, Saarland Univ., doi:10.22028/D291-23134                            | <b>The source of §11.10.4's tree rules.</b> Open access, German, now in <a href="#">APhi_Papers/</a> as <a href="#">Klis2015_PhDThesis_FastFEM_MagnetoQuasistatic_DE.pdf</a> . Uses the <b>A-V-A formulation</b> — i.e. an A- $\Phi$ method, not an E-field one, so its tree treatment should transfer to us more directly than Stysch's does. |
| <b>Dyczij-Edlinger &amp; Bíró (1996)</b> , <i>A joint vector and scalar potential formulation for driven high frequency problems using hybrid edge and nodal finite elements</i> , IEEE T-MTT <b>44</b> (1) | cited <a href="#">[DB96]</a> for **gauged <i>and</i> ungauged non-lossy potential formulations**. "Joint vector and scalar potential" with "hybrid edge and nodal finite elements" is <i>exactly our discretization</i> , which makes this the most directly comparable prior formulation found so far. Not obtained.                          |
| <b>Hiptmair, Kramer &amp; Ostrowski (2008)</b>                                                                                                                                                              | already in §11.6; now known to be one of Eller's two parents, and to give singular non-symmetric matrices.                                                                                                                                                                                                                                     |
| <b>Jochum et al. (2015)</b> and <b>Joc+16</b>                                                                                                                                                               | already in §11.6; now known to be the other parent, PEC-equivalent to Eller, and expensive through matrix fill.                                                                                                                                                                                                                                |
| <b>Eller (2017)</b> , <i>A Low-Frequency Stable Maxwell Formulation in Frequency Domain and Industrial Applications</i> , dissertation, TU Darmstadt, tubiblio/92810                                        | the long form of the SIAM paper. <b>Record only — no full text on TUpriints</b> , and per the thesis it omits the tree-creation details too, so it is not worth chasing either.                                                                                                                                                                |
| <b>Kruskal (1956)</b>                                                                                                                                                                                       | the spanning-tree algorithm they use.                                                                                                                                                                                                                                                                                                          |

## 12. Zhao & Fu's Coulomb gauge — what Maxwell shipped, and why it half-worked

Three Zhao & Fu papers, all IEEE Trans. Magn. **53**(6), June 2017, have been in [APhi\\_Papers/](#) throughout. Only the frequency-domain one had been read, and it is summarised in the bibliography in a single line. **None of the three had been read for the**

**mixed-material port question**, which is the one that matters — these are the papers Maxwell's mixed-material Coulomb gauge is built on.

#### FILE

|                                                          |                                                                                                 |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <a href="#">AphiFreqDomainCoulombgauge_yanpu2017.pdf</a> | A New Stable Full-Wave Maxwell Solver for All Frequencies. $\chi$ over the <b>whole</b> domain. |
| <a href="#">AphiTdomainColoumbgauge_yanpu2017.pdf</a>    | time domain. $\chi$ in the <b>non-conducting region only</b> .                                  |
| <a href="#">AphiMstaticColoumbgauge_yanpu2017.pdf</a>    | magnetostatic. $\chi$ over the whole domain.                                                    |

## 12.1 The formulation

A nodal **dummy scalar**  $\chi$  imposes the Coulomb gauge on an edge-element  $A$ .

Frequency domain:

$$\begin{aligned}\nabla \times (\nabla \times A) + (j\omega\sigma - \omega^2\epsilon)A + (\sigma + j\omega\epsilon)\nabla\varphi - \nabla\chi &= Js \\ -\nabla \cdot [ (j\omega\sigma - \omega^2\epsilon)A + (\sigma + j\omega\epsilon)\nabla\varphi ] &= 0 \\ \nabla \cdot A + \chi &= 0\end{aligned}$$

giving a 3×3 block system in  $(A, \varphi, \chi)$  with  $K_{A\chi} = K_{A\varphi} = \int N_i \cdot \nabla N_j$  and  $K_{\chi A} = K_{\varphi A} = \int \nabla N_i \cdot N_j$ , i.e.  $\chi$  couples exactly like  $\varphi$  but with coefficient 1 instead of  $(\sigma + j\omega\epsilon)$ . The printed system is unsymmetric — the (2,1) block is  $j\omega$  times the transpose of the (1,2) — and the  $-j/\omega$  factor in their text is the row scaling that symmetrises it. Same device as Balian's variant (ii)/(iv) (§11.1).

**Why a dummy scalar at all** — their own statement of the obstacle, and it is the cleanest in the collection:

*"Since the divergence of an edge element basis function is zero within each mesh element, the use of the gauged formulation proposed in [4] and edge element for the MVP will result in the curl–curl equation." "A piecewise divergence-free function may have a globally large weak divergence because of the jumps of its normal component at the interior faces between elements. Therefore, **the global divergence of the MVP represented by edge element is not necessarily 0.**"*

So the penalty method is unusable on Whitney-1 (this is the same  $\nabla \cdot \omega_{mn} \equiv 0$  fact that forces Chew's intermediate variable  $d$ , §5 of [GENERALIZED\\_LORENZ\\_GAUGE.md](#)), and  $\chi$  controls the *global weak* divergence instead.

In the time-domain paper the gauge is imposed **only in  $\Omega_n$** , and the uniqueness argument is worth stating because §12.3 turns on it: take **div** of the first equation, find that  $\chi$  satisfies **Laplace's equation** in  $\Omega_n$ , then impose **homogeneous Dirichlet  $\chi = 0$**  on  $\partial\Omega_n$  — whence  **$\chi \equiv 0$** , and the second equation reduces to the Coulomb gauge.

## 12.2 Why it is a real improvement at a material interface

Their motivation is precisely our §6:

"since the nodal elements impose both tangential and normal continuity of  $A$ , which has tangential continuity only across material interfaces, it can produce **large errors at material interfaces or re-entrant corners** of problem domains." "edge elements ... allow **necessary discontinuity in the normal component** of the MVP."\*

That is the whole point of the construction: keep the normal jump that Whitney-1 permits, and still gauge  $A$ . Against a *nodal* Coulomb gauge it is a genuine advance, which is consistent with your recollection that Maxwell's Coulomb option "was doing some improvement."

## 12.3 Why it would not fully work at a mixed-material port

Three reasons. The third is a fact about the papers; the first two are inference and should be read as such.

**1. The gauge constraint is  $\epsilon$ -blind.**  $\chi$  enforces  **$\nabla \cdot A = 0$**  in a global weak sense. But the correct interface condition at an  $\epsilon$  discontinuity is on  **$\epsilon A$** , not  $A$  — Chew's (26),  **$\hat{n} \cdot (\epsilon_1 A_1) = \hat{n} \cdot (\epsilon_2 A_2)$**  (§2.2, §6). Edge elements *permit* the normal jump; the Coulomb constraint carries no  $\epsilon$  and therefore says nothing about **how large the jump should be**. Chew's generalized gauge builds the  $\epsilon$ -weighted condition into the PDE itself; Coulomb cannot, at any order of element. This is the principled reason to expect partial success, and it is the same conclusion §6.3.1 reached from Ansari.

**\*\*2.** The  **$\chi \equiv 0$**  argument needs a boundary that an interior mixed port does not provide.  
 **$\chi \equiv 0$**  follows from Laplace in  $\Omega_n$  *plus*  **$\chi = 0$**  on all of  $\partial\Omega_n$ . A port sheet interior to the dielectric — the coax annulus of **COAX\_PORT\_ANALYSIS.md** — is not part of  $\partial\Omega_n$ . So either  $\chi$  is pinned on it, which imposes an extra, unphysical constraint on  **$\nabla \cdot A$**  across the port, or it is not, and the uniqueness argument no longer closes, leaving  **$\chi \neq 0$**  to pollute the solution. Either branch predicts "improved but not right". **This is testable** and is the sharpest experiment this section suggests: solve a mixed port with  $\chi$  pinned on the port sheet and with it free, and compare.

### 3. It was never demonstrated on the case. The validation across the three papers is:

|                  |                                                                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| frequency domain | single copper conductor bar; parallel-plate capacitor; power inductor (against COMSOL 5.2a); two copper conductors                                                                                    |
| time domain      | TEAM Workshop Problem 7 (racetrack coil over an aluminium plate with a hole, 50 Hz); a second TEAM benchmark; eddy currents in a helical copper conductor — compared against <b>ANSYS Maxwell 3-D</b> |
| magnetostatic    | linear and nonlinear magnetostatic examples                                                                                                                                                           |

**Not one mixed-material port.** Eddy-current benchmarks, one plain capacitor, one inductor. And the port model is conductor-only by construction: their voltage excitation sets  $\phi$  takes value of 0 at one terminal and 1 at the other terminal of the **solid conductor** — which matches your recollection that Maxwell restricted floating potentials to ports intersecting a conductor. The formulation was shipped against a case it had never been shown to handle.

#### 12.4 Two things these papers settle for us

**They are a published statement that the tree matters.** The time-domain paper, on tree–cotree:

*"the choice of the tree **affects significantly the accuracy** of the approximation. The use of an arbitrary tree results in poor convergence."*

The Bibliography below has claimed that no paper in this collection says tree–cotree costs anything, with Munteanu's aside about **A** as the nearest approach. **That claim was too strong** and is corrected there. This is still about **A** and accuracy rather than about  $\phi$  specifically, so **07\_GaugeInvariance**'s measurement of  $\phi$  remains our own result — but it is no longer an isolated one, and Zhao & Fu cite two further references for it.

**They contradict Herles et al. on the fake-conductivity trick:**

*"The small fake conductivity regularization method ... involves an artificial conductivity parameter in the non-conducting region, which **may cause unexpected errors in practice** when using this formulation."*

Herles et al. (§11.2) use exactly that,  $\sigma_{art} = 1e-6$ , inside their scaling weight  $\beta$ . A direct disagreement in the literature, and one to keep in view if we adopt their scaling.

### 12.5 Where this leaves the gauge ranking

Zhao & Fu is the **best-documented Coulomb-family option**: symmetric after one row scaling, direct-solvable, edge elements for  $A$  with nodal  $\varphi$  — our discretization — and validated to 1.38 M unknowns. If the goal were a conductor-port eddy-current solver it would be the cheapest credible route.

It does not reach the mixed port, for the reasons above, and §12.3's reason 1 is structural rather than a matter of implementation effort. That keeps the ranking of §8 as it stands:  
\*\*Coulomb-family gauges improve the interface behaviour without resolving it, because the condition they impose does not carry  $\epsilon$ .\*\*

Your own observation — that Maxwell's Coulomb option improved matters without fixing them — is now matched to a mechanism, and §12.3's reason 2 gives a cheap experiment that would distinguish mechanism 1 from mechanism 2.

---

## Bibliography

Papers obtained and read for this edition, all now in [APhi\\_Papers/](#) :

|                                                        | WHAT IT CONTRIBUTES                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chew (2014)</b> , PIER <b>149</b> , 69–84           | The generalized gauge $\nabla \cdot (\epsilon A) = -\chi \partial \Phi / \partial t$ , $\chi = \alpha \epsilon^2 \mu$ ; interface conditions embedded in the PDEs; generalized Green's theorem, extinction theorem, equivalence principle and SIEs for $A$ - $\Phi$ . <b>The keystone.</b> |
| <b>Sharma &amp; Triverio (2022)</b> , arXiv:2112.07360 | Modified Lorenz gauge inside lossy conductors; explicit statement that the $\Phi$ reference contaminates $\nabla \cdot A$ . DC to high frequency, surface-only conductor modelling.                                                                                                        |
| <b>Sharma &amp; Triverio (2021)</b> , arXiv:2108.02764 | Companion: potential-based BEM for lossy materials.                                                                                                                                                                                                                                        |
| <b>Abdrabou &amp; Gomez (2026)</b> , arXiv:2507.02099  | DEC + SIE under Lorenz gauge; no discrete gauging because $A$ is three 0-forms; 14→2 operator reduction; free-space coupling-surface constraint.                                                                                                                                           |
| <b>Li, Fu &amp; Shanker (2017)</b> , arXiv:1705.00265  | Potential integral equations, Lorenz gauge, decoupling gives low-frequency stability; second-kind, well-conditioned operators.                                                                                                                                                             |

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSARI,<br>FARQUHARSON &<br>MACLACHLAN (2017),<br>GJI 210(1), 105–129   | EXPLICIT LAGRANGE-MULTIPLIER COULOMB GAUGE. DEMONSTRATES NON-<br>UNIQUE POTENTIALS BY SHOWING DIRECT AND ITERATIVE SOLVERS DISAGREE<br>ON $\Phi$ AND A WHILE AGREEING ON THE FIELDS. USES EDGE ELEMENTS FOR <b>A</b><br>AND NODAL FOR $\Phi$ — OUR DISCRETIZATION. IDENTIFIES THE CAUSE AS $\hat{\mathbf{n}} \cdot \mathbf{A}$<br>BEING DISCONTINUOUS ACROSS INTERFACES WHILE THE COULOMB GAUGE<br>REQUIRES IT CONTINUOUS (§6.3.1). |
| Rapetti, Alonso<br>Rodríguez & De Los<br>Santos (2022), J5(1),<br>52–63 | High-order tree–cotree in general domains. States outright that <b>tree gauges are not a discretization of the Coulomb gauge and enforce no orthogonality</b> , and that the tree degrees of freedom are <b>set arbitrarily</b> without affecting <b>B</b> . The structural reason tree-gauged $\Phi$ has no continuum limit.                                                                                                       |

Previously read, retained from the first edition:

|                                                         |                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Munteanu, <i>Tree-cotree condensation properties</i>    | Variants A–E and their conditioning. §IV: non-uniqueness of <b>A</b> is "in principle irrelevant"; gauging exists for regularity. §V.E: tree-independence possible, costs sparsity. <b>A-only, magnetostatic — contains no <math>\Phi</math> and cannot speak to its uniqueness.</b>                                                                     |
| Li, Sun, Dai & Chew (2015), IEEE Trans. Magn.           | Generalized Coulomb gauge (static limit of the Lorenz one). Ungauged systems let iterative solvers converge to different answers; constant gradients live in the edge space.                                                                                                                                                                             |
| Li, Sun, Dai & Chew (2016), IEEE TAP 64(10)             | FEM implementation of the generalized-Lorenz A- $\Phi$ formulation. Two null-space-free Helmholtz equations; conditioning $\sim 1e6$ against $1e17$ . Needs Hodge operators and a sparse approximate inverse.                                                                                                                                            |
| Zhao & Fu (2017), IEEE Trans. Magn. 53(6), three papers | Coulomb gauge via a dummy scalar proven zero; symmetric after one row scaling, direct-solvable, 1.38 M unknowns. <b>The basis of Maxwell's mixed-material Coulomb option — now read properly in §12</b> , including why it improves interface behaviour without resolving it and the fact that none of the three papers validates a mixed-material port. |
| Lee & Jin (2008), MOTL 50(6)                            | Tree–cotree conditioning; splitting is not unique; root choice matters; conditioning still degrades at low frequency.                                                                                                                                                                                                                                    |
| Su Yan (2021), PIER M 106                               | Auxiliary scalar potential to eliminate tree–cotree graph searches while keeping all-frequency stability.                                                                                                                                                                                                                                                |

**No paper in this collection states that tree–cotree yields a non-unique  $\Phi$**  — but, per §12.4, Zhao & Fu (2017) do state that "*the choice of the tree affects significantly the accuracy of the approximation*" and that "*the use of an arbitrary tree results in poor convergence*", citing two further references. That is about **A** and accuracy rather than about  $\Phi$ 's uniqueness, so the distinction below still holds, but it is a much closer published

statement than this paragraph previously allowed. The nearest are Munteanu's tree-dependence remark (about **A**) and Sharma & Triverio's reference-dependence of  $\nabla \cdot \mathbf{A}$ . Our 15 % measurement in `07_GaugeInvariance` remains our own result and should keep being presented as such.